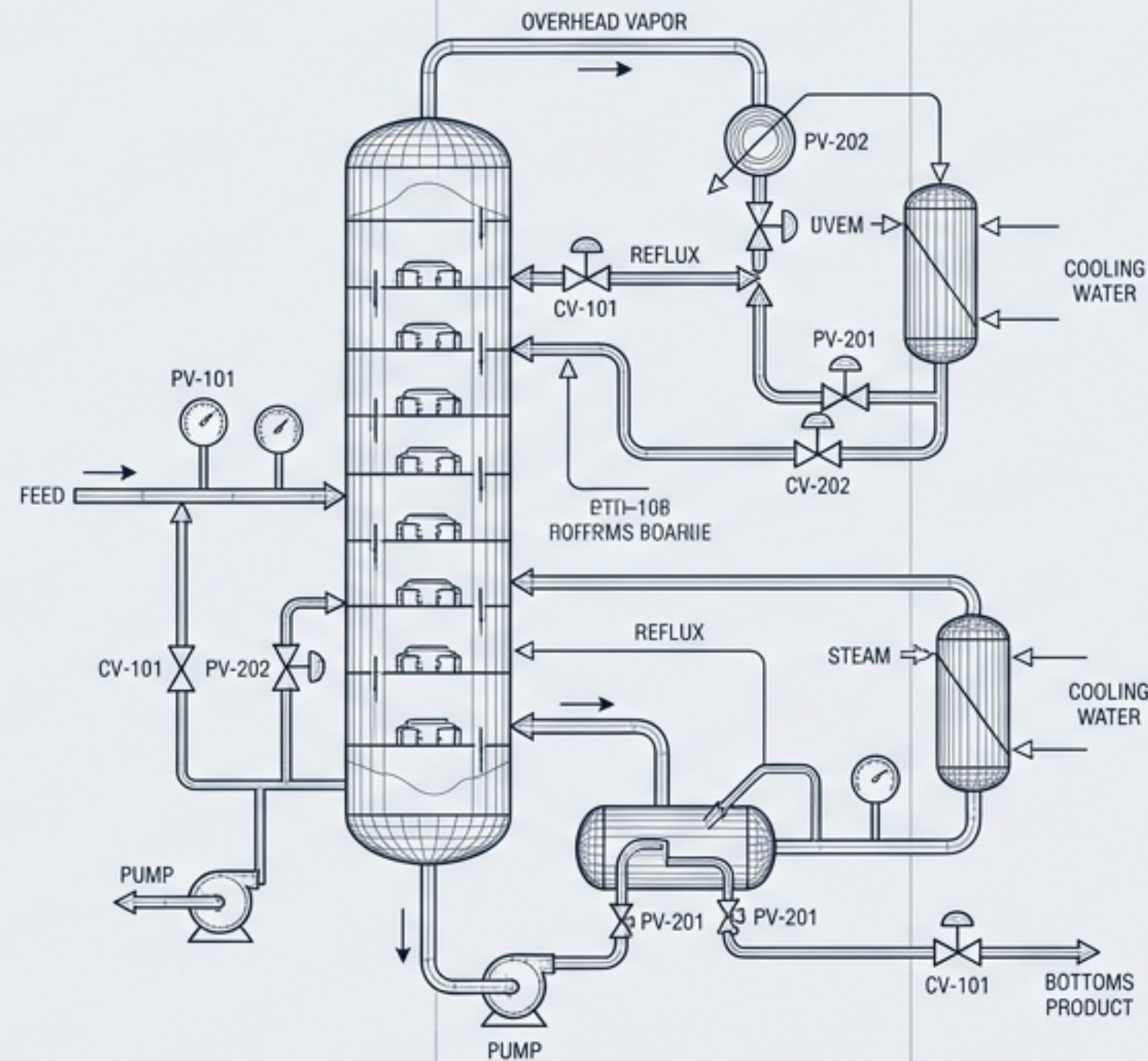


From Documents to Decisions

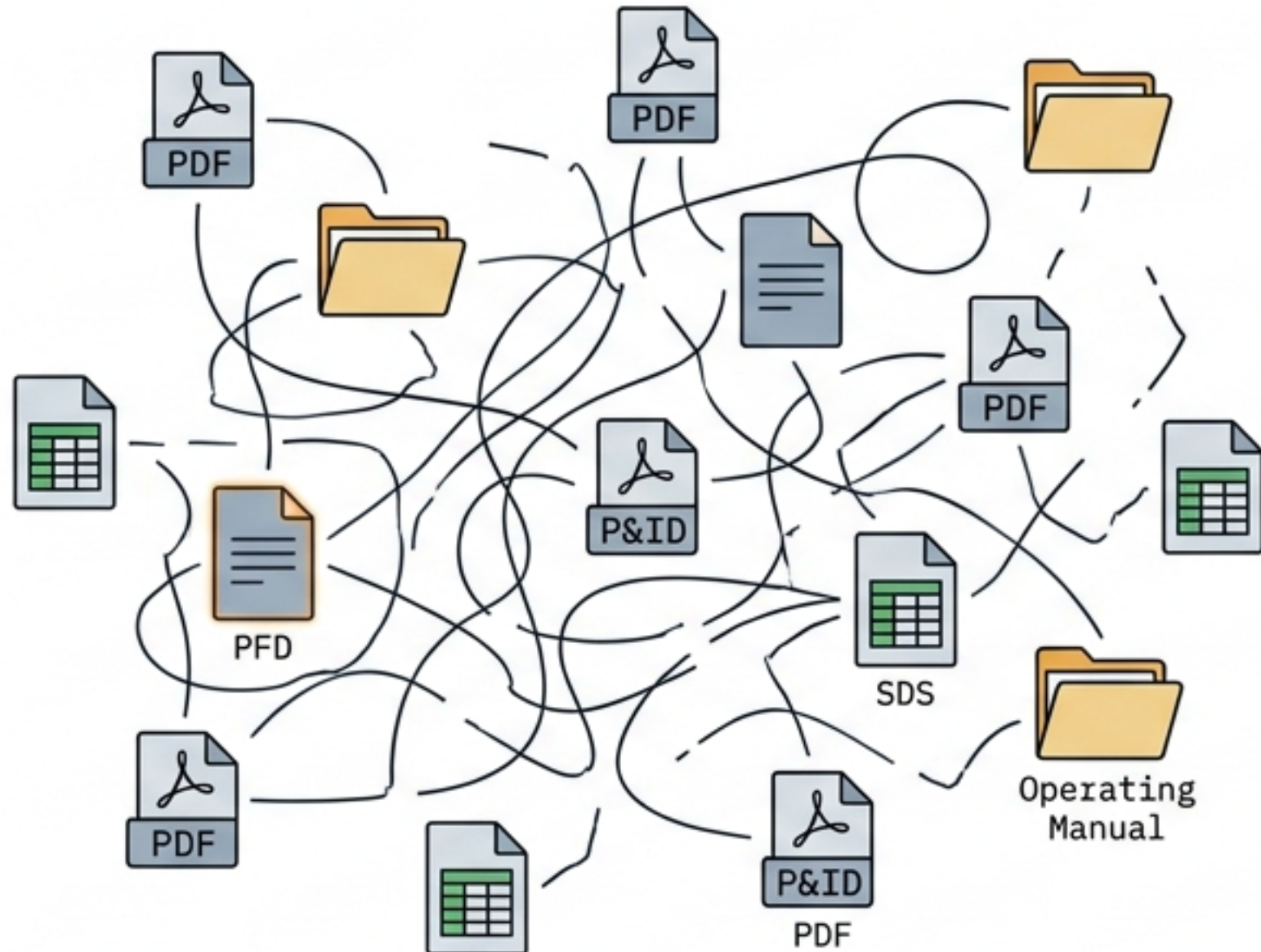
Building a Compounding Process Knowledge System with LLM Wiki + Claude Code



Proof Case: PTT Phenol Train II (PPCL) — UOP Process — CDN Section HAZOP

The Entropy of Process Knowledge

The Web of Chaos



The Symptom

**Where is the
design pressure
for E-2303?**

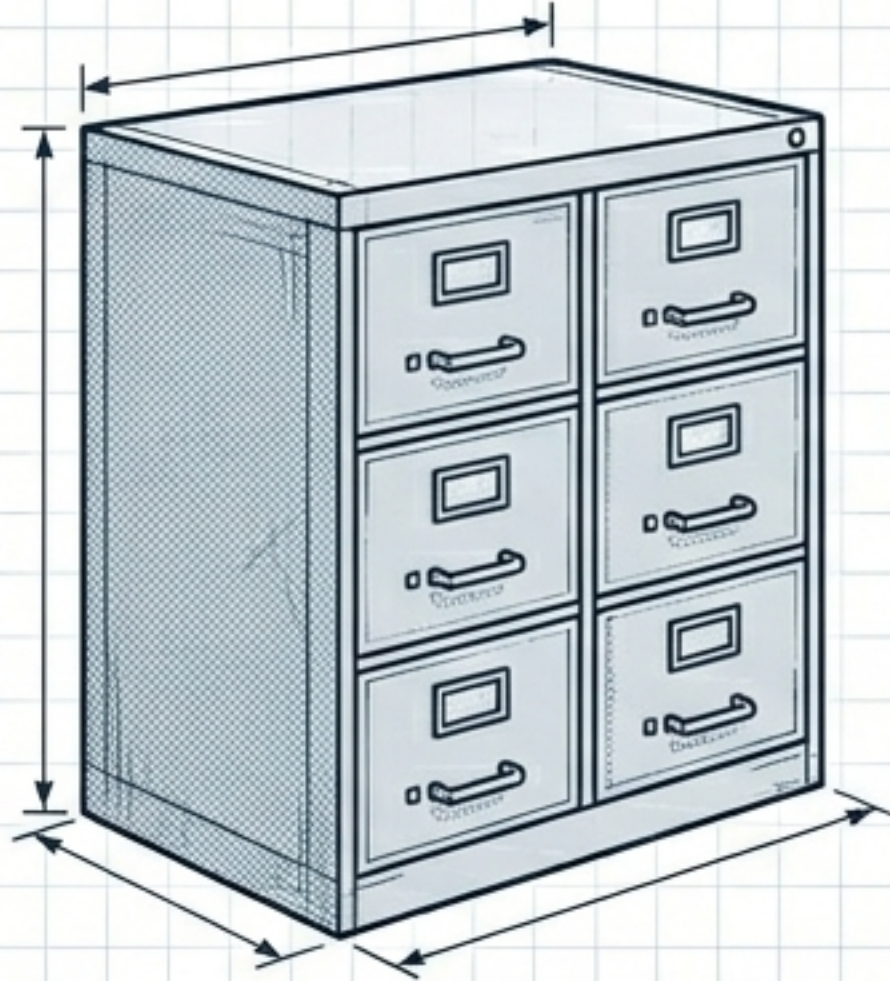
■ Answers do not accumulate;
every search starts from zero.

● Tribal knowledge walks
out the door.

◆ The result: 20 minutes of
PDF hunting per question.

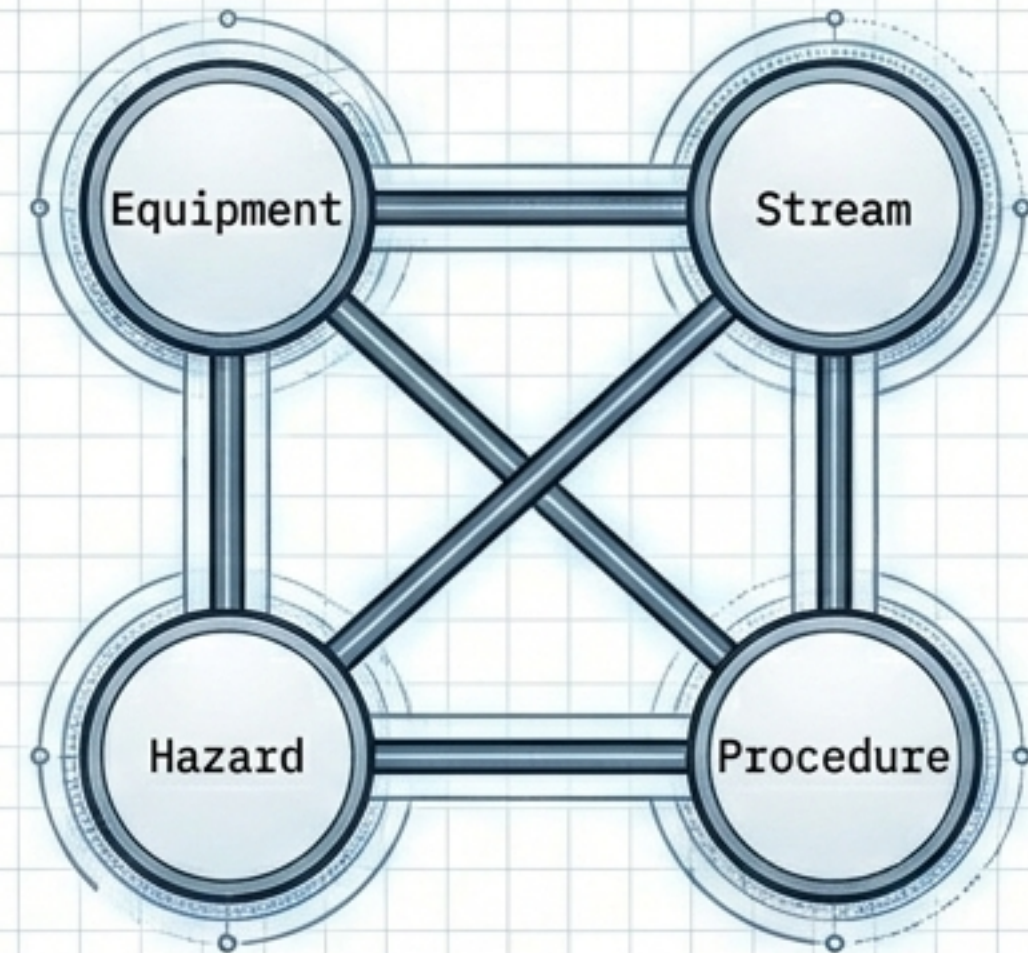
The Knowledge Shift: From Storage to Synthesis

Old Way



On-Demand Retrieval

New Way



Maintained Synthesis

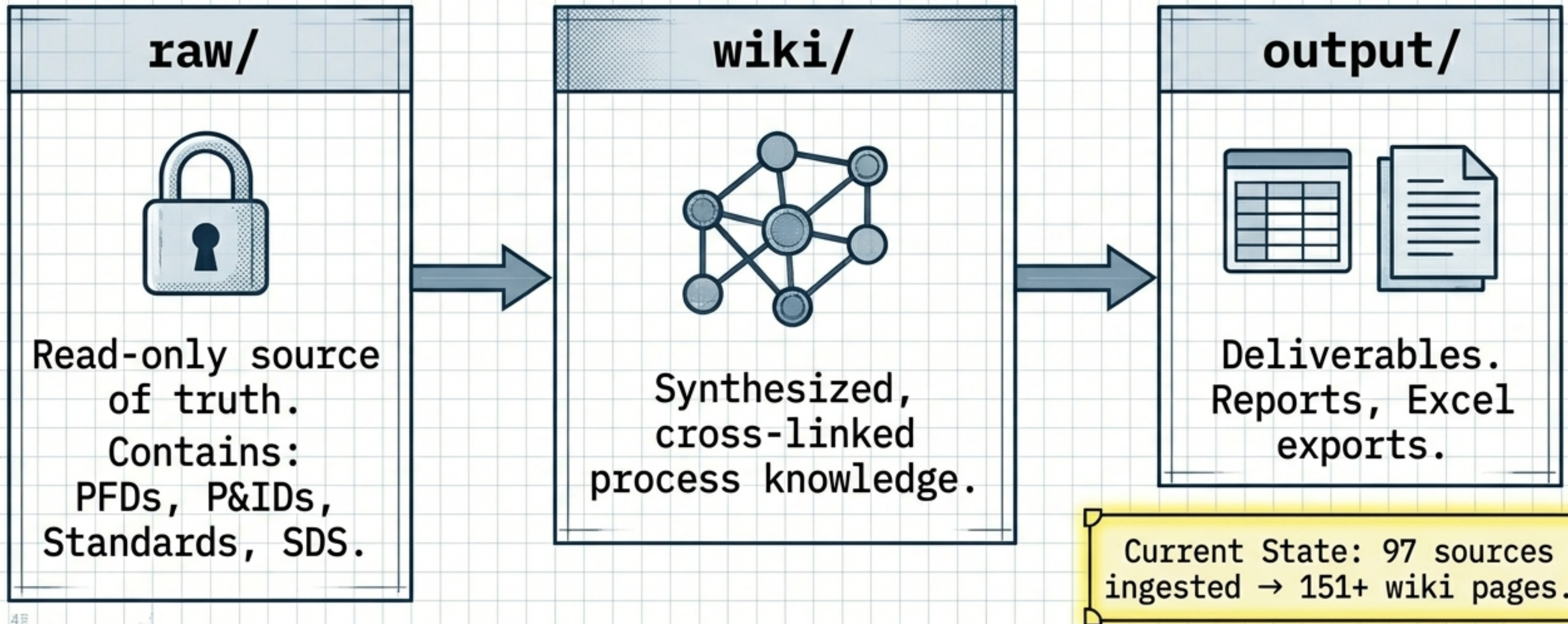
What if every document made the next answer better? The system behaves like a colleague who has read and remembers all 97 documents.

Defining the LLM Wiki Paradigm

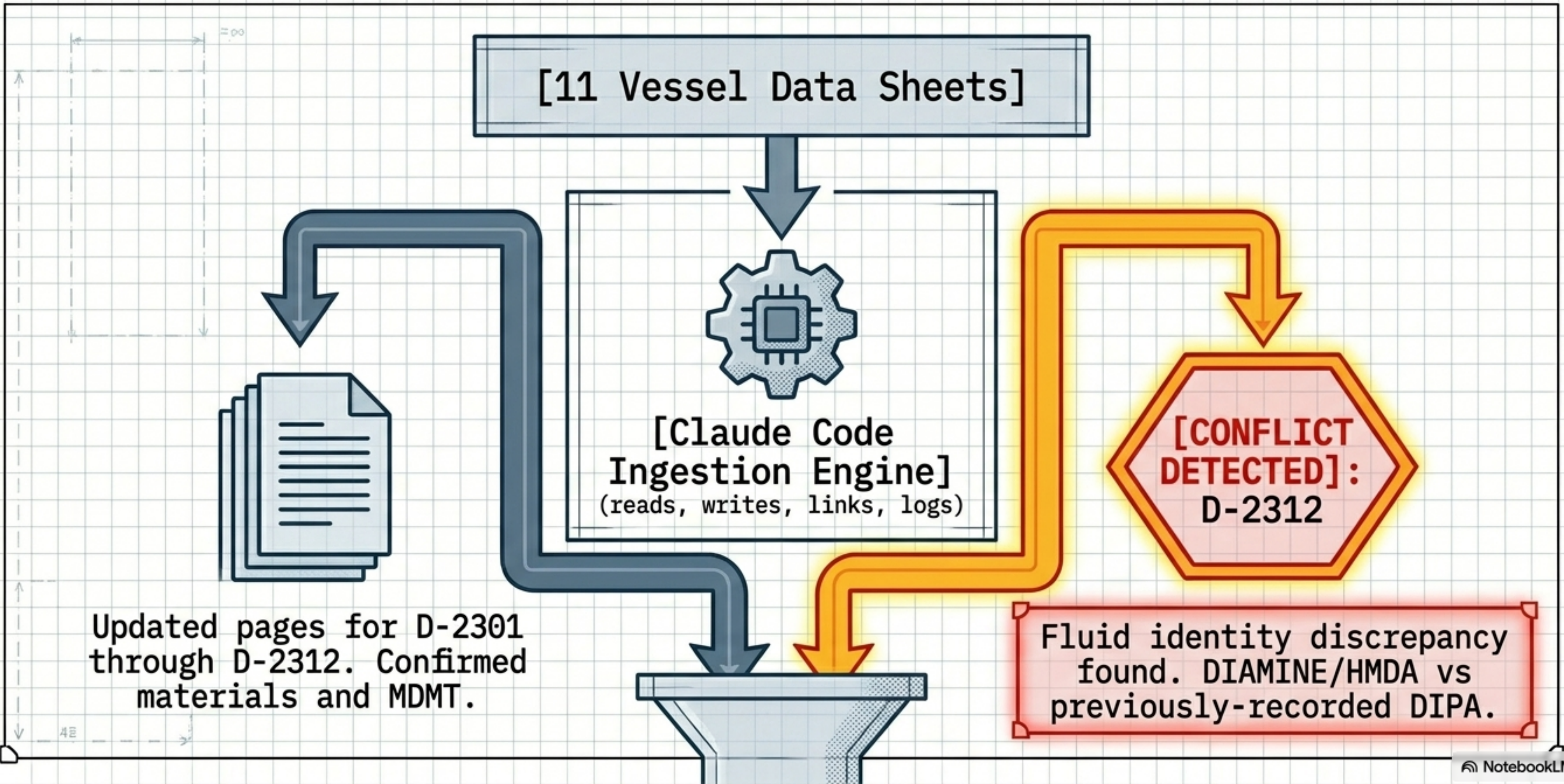
	Standard Folder & Search	Standard RAG Chatbot	LLM Wiki + Claude Code
Operation	Manual Retrieval	Chunk Indexing	Synthesized Markdown Base
Knowledge State	Static & Disconnected	Point-in-time answers	Compounding & Cross-Referenced
Contradictions	Hidden in files	AI Hallucinates / Averages	Explicitly Flagged: [CONFLICT]
Auditability	Nil	Medium (quotes chunks)	High (Read-only Raw vs. Maintained Wiki)

The Knowledge Engine Architecture

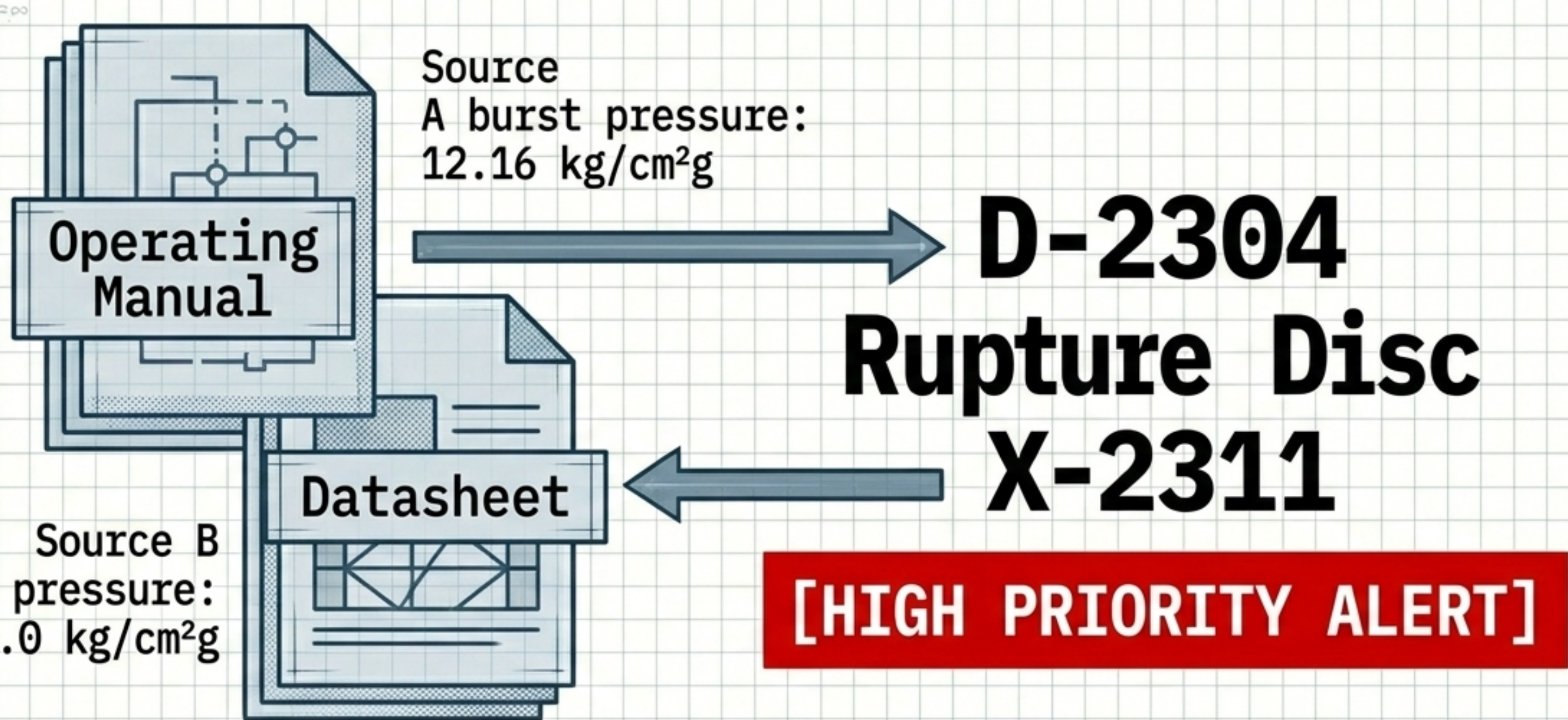
CLAUDE.md Schema (Enforces rules every session)



Active Ingestion & Auditing



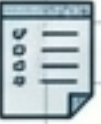
Real-World Conflict Resolution



System flagged contradiction on the sole overpressure protection for the highest-hazard vessel in the section. Caught before HAZOP begins.

Division of Labor: Why HAZOP?

The Machine's Domain (80%)

- ➔  • Automating tedious data assembly.
-  • Sourcing design pressures. ———→
-  • Pulling stream compositions. ———→
-  • Formatting worksheets. ———→
-  • Checking relief devices. ———→

The Human's Domain (20%)

-  • Preserving expert judgment.
-  • Setting node boundaries.
-  • Assessing qualitative risk.
-  • Determining safeguard validity.

Why HAZOP? High-stakes, structured, repetitive, and deeply standards-driven.
If the agent is grounded enough for HAZOP, it works for anything.

Building the HAZOP Skill Stack

The Active Variable

Marked-up P&ID

Company Standards

P- (Q-MP) -OEMS-005 R4
(HAZOP Procedure)

W- (Q-MP) -002 R2
(Risk Matrix)

SG- (Q-MP) -014 R3
(9-Step Method)

Synthesized Process Base

Wiki equipment data, 15 GHS SDS, Cause & Effect tables.

The agent executes a fixed workflow with a rigid node worksheet schema and three standardized risk blocks.

The Three Trust Guardrails

Filter 1: Anti-Bias



Prior reports are
NEVER ingested during
an active study.
Prevents anchoring.

Filter 2: Standards Primacy



Company documents
strictly govern.
General knowledge
gets a [VERIFY] flag.

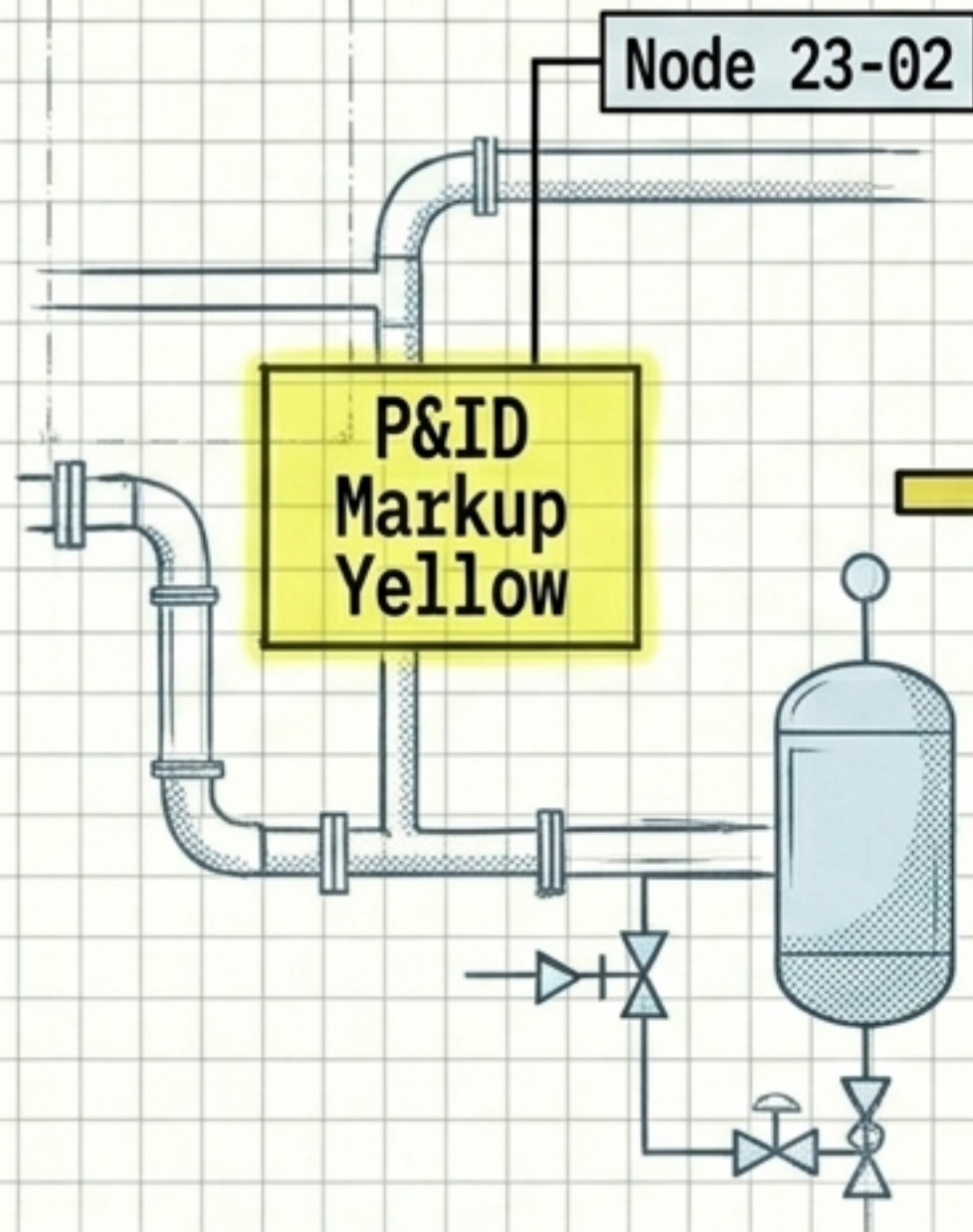
Filter 3: Node Boundary



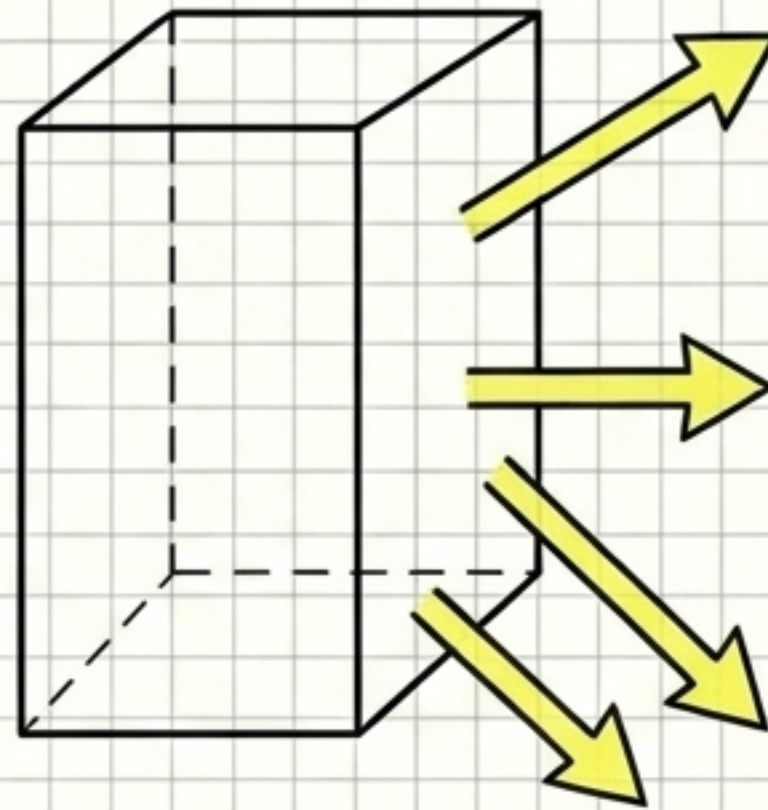
AI reads the
engineer's markup.
It **NEVER** invents
its own scope.

The Human-Machine Interface

1. The Engineer's Boundary



2. The System's Interpretation



The AI reads the markup, extracting exact equipment boundaries without scope creep.

3. The Parameter Pull

Tag	Temp (°C)	Press (barg)
E-2302	150	20
P-2308	45	8
V-2300	120	5

The AI automatically queries the wiki, pulling operating parameters itemized per equipment tag to confirm design intent.

Deliverables Arsenal: CDN Section

Real Plant Executions

CDN-N02

Preflash Column feed-heating circuit (E-2302A/B). Dominant hazard correctly grounded: CHP over-temp above $\sim 80^{\circ}\text{C}$ onset. 13 deviations tracked.

CDN-N03

Flash Column Vaporizer (E-2304). 17 deviation chains tracked.

The Artifacts



EXCEL
WORKSHEET



REPORT
DOCUMENT



CHECKLIST

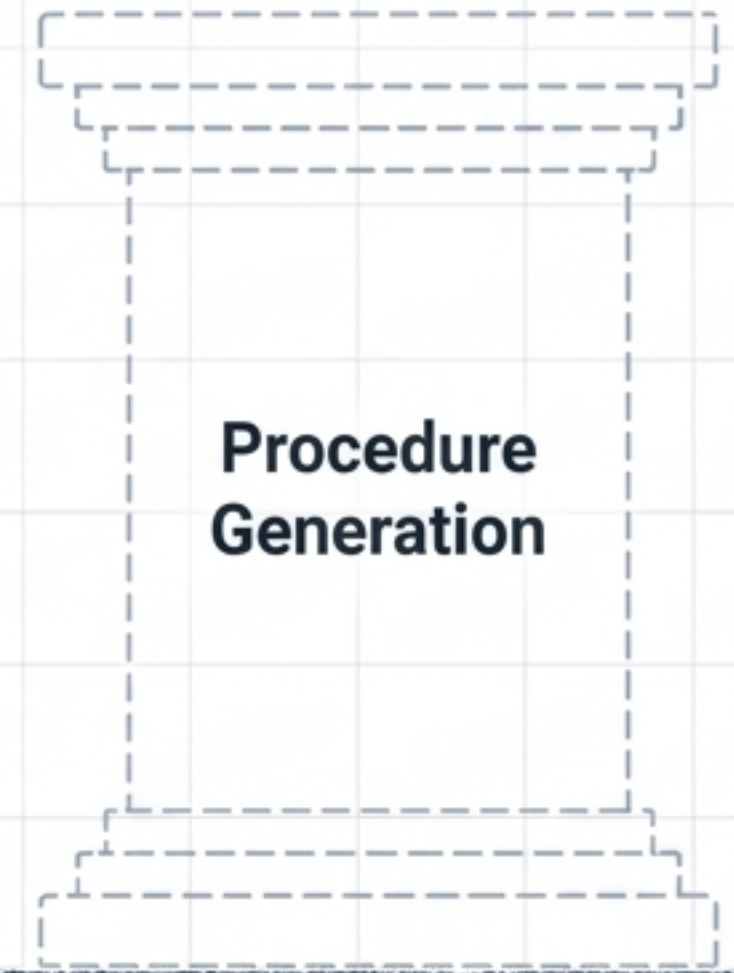
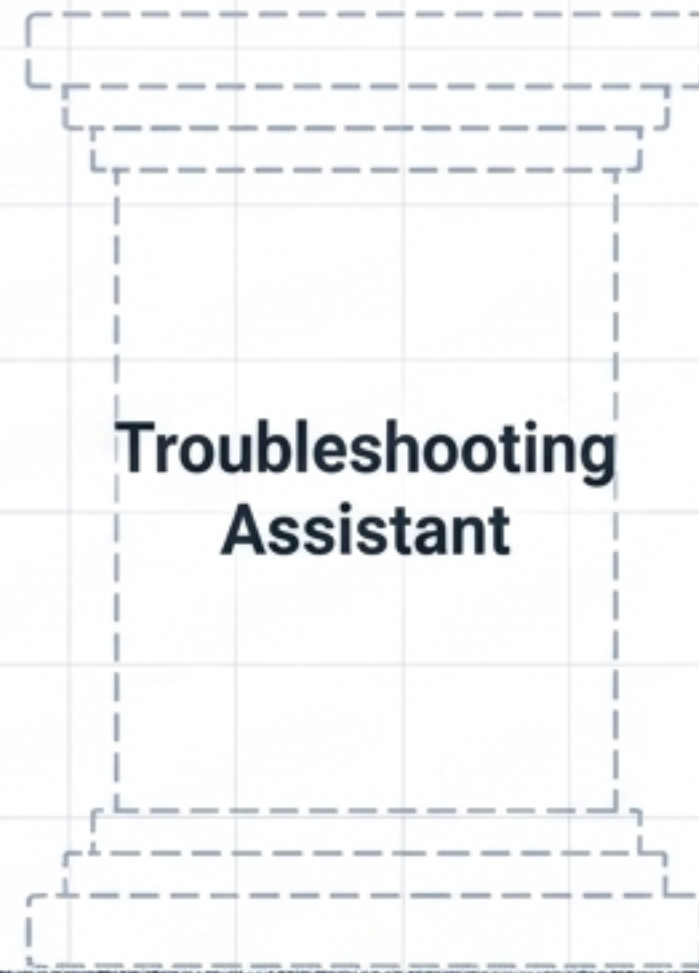
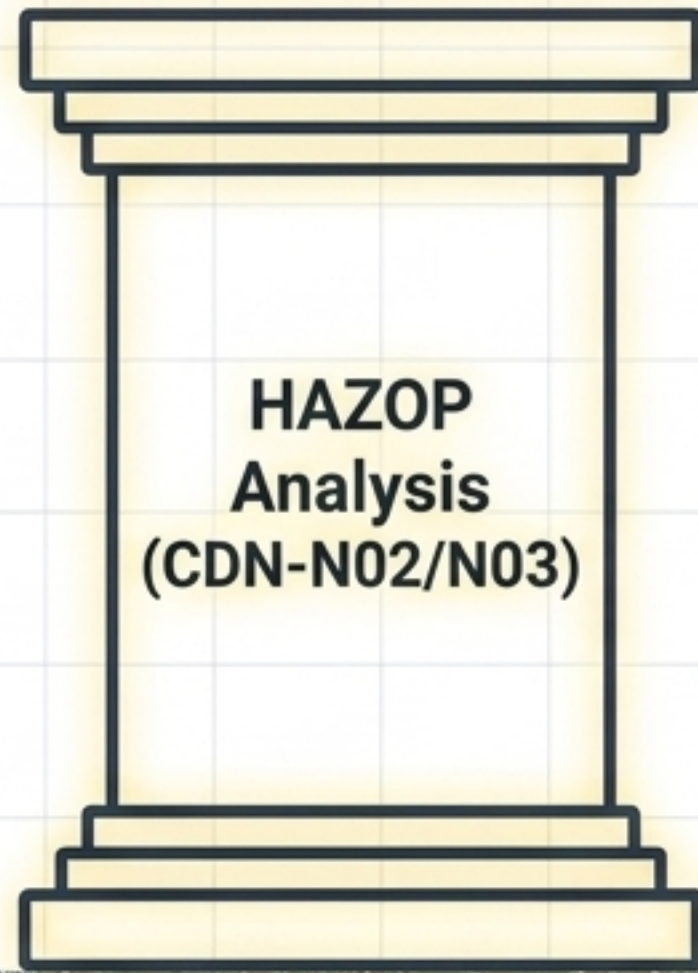
Every deviation, safeguard (with IPL credit), and recommendation is directly traceable to a source page.

Live Validation Checklist

Target Node:
Node 23-05 / 23-06
(Cumene Quench Drum
area markup)

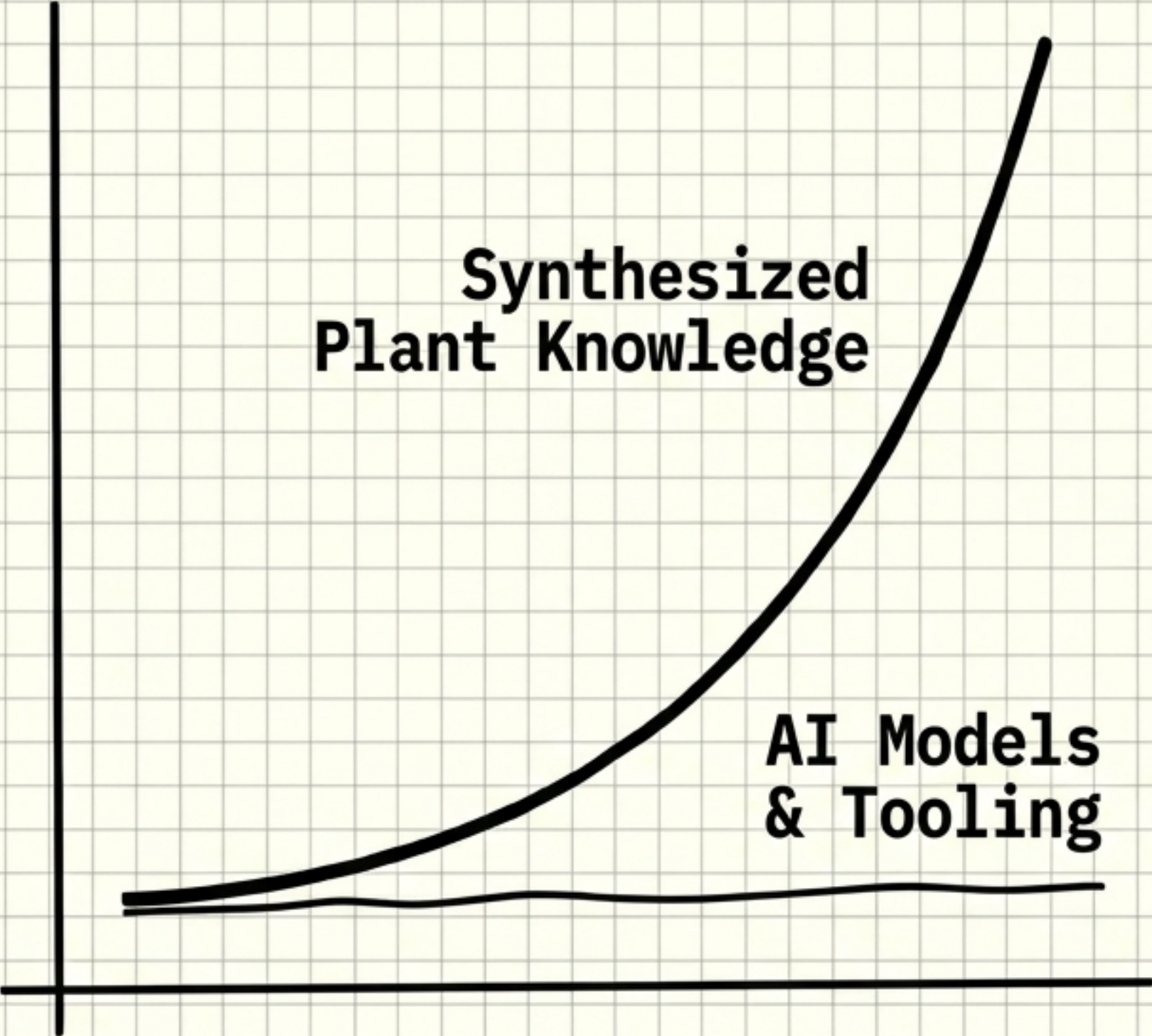
- 1.** **Scope:** Does it respect the marked boundaries exactly?
- 2.** **Sourcing:** Are parameters explicitly cited from the wiki, not invented?
- 3.** **Grounding:** Are deviations rooted in real equipment and plant chemistry?
- 4.** **Ranking:** Are recommendations traceable and ranked strictly per our W-(Q-MP)-002 R2 matrix?

Because we built the knowledge base once, every subsequent tool is cheap, fast, and completely grounded.



LLM WIKI (Validated, Cross-Linked Process Knowledge)

The Moat: Compounding Knowledge



The graph features a vertical y-axis and a horizontal x-axis. Two curves originate from the same point on the x-axis. The upper curve, labeled 'Synthesized Plant Knowledge', rises steeply in an exponential fashion. The lower curve, labeled 'AI Models & Tooling', remains relatively flat and close to the x-axis throughout the visible range.

Synthesized
Plant Knowledge

AI Models
& Tooling

- The Wiki is the asset – the moat.
- Every document ingested permanently raises the baseline for all future use cases.
- Human judgment remains central; the system eliminates document archaeology so engineers can focus on engineering.

We built the foundation. What would you point it at?

1. Feed It

Contribute the source documents you trust most.

2. Challenge It

Review the CDN-N02/N03 outputs and find the gaps.

3. Direct It

Propose the next agent for your highest-friction workflow.